



JERUSALEM COLLEGE OF ENGINEERING

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

QUESTION BANK

Regulation	:	2023
Course Name	:	DATABASE MANAGEMENT SYSTEMS
Course Code	:	JCS2401
Year / Semester / Branch		II/IV/CSE
Unit No	:	1
Unit Title	:	DATA MODELS AND QUERYING
No. of Hours	:	9
Weightage in Exam (%)	:	20 %

PART A

Q.No.	Questions	Marks	BTL	CO
1.	Define database management system Database management system (DBMS) is a collection of interrelated data and a set of programs to access those data.	2	K1	CO1
2.	List any five applications of DBMS. • Banking, • Airlines • Universities, • Tele communication, • Finance, Sales, Manufacturing • Human resources	2	K1	CO1
3.	Outline the purpose of a DBMS. Data redundancy and inconsistency, Difficulty in accessing data, Data isolation, Integrity problems, Atomicity problems and Concurrent access anomalies.	2	K2	CO1
4.	Define instance and schema Instance: Collection of data stored in the data base at a particular moment is called an Instance of the database Schema: The overall design of the data base is called the data base schema.	2	K1	CO1
5.	Define the terms physical schema and logical schema. Physical schema:			



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	The physical schema describes the database design at the physical level, which is the lowest level of abstraction describing how the data are stored. Logical schema: The logical schema describes the database design at the logical level, which describes what data are stored in the database and what relationship exists among the data.	2	K1	CO1
6.	Classify the different types of data models used in database systems. • Hierarchical model • Network model • Relational model • Entity-Relationship (ER) model • Object-oriented model..	2	K2	CO1
7.	Outline the operations of relational algebra. Relational algebra is a procedural query language used to perform operations on relations (tables) in a database. It provides a set of fundamental operations to retrieve and manipulate data. These operations include selection, projection, union, set difference, and join. Relational algebra forms the theoretical foundation of SQL. It helps in understanding how queries are processed internally. It is important for designing efficient database queries and operations.	2	K2	CO1
8.	Categorize and define the different views of data in a database system. A database system has three levels of data abstraction: physical, logical, and view levels. The physical level describes how data is stored in storage devices. The logical level explains what data is stored and the relationships among data. The view level defines how users interact with the data. This separation helps in data independence and simplifies database management. Each user can have a customized view of the same database.	2	K2	CO1
9.	Why is a data dictionary important in DBMS? A data dictionary in DBMS is important because: • It stores metadata (data about data) such as table structure, data types, and constraints. • It ensures data consistency, integrity, and helps in efficient database management.	2	K2	CO1
10.	Compare SQL with relational algebra	2	K2	CO1



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	Feature	SQL (Structured Query Language)	Relational Algebra			
	Type of Language	Declarative (what to do)	Procedural (how to do)			
	User Friendliness	Easy and user-friendly	Less user-friendly			
	Purpose	Used for querying and managing databases	Theoretical foundation for DBMS			
	Usage	Used in real-world applications	Mainly used for understanding/query design			
	Execution	Directly executed in DBMS	Not directly executed			
	Level	High-level language	Low-level formal language			
	Operations	Uses SELECT, INSERT, UPDATE, DELETE	Uses Select, Project, Join, etc.			
11.	Differentiate between the database approach and the file-based approach based on their characteristics. File -based System. • Separation and isolation of data • Duplication of data • Incompatible file formats • Data dependence Database Approach: • Control of data redundancy • Data consistency • Sharing of data • Improved data integrity • Improved security			2	K2	CO1
12.	Name the disadvantages of file processing system? The file processing system has the following major disadvantages: □ Data redundancy and inconsistency □ Integrity Problems □ Security Problems			2	K1	CO1



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	□ Difficulty in accessing data □ Data isolation.											
13.	Interpret the features of a query language in DBMS. A query language is a language used to communicate with a database to retrieve and manipulate data. It allows users to perform operations such as inserting, updating, deleting, and retrieving data. Query languages provide a structured way to interact with databases without needing to know internal details. They help in efficient data access and management. The most commonly used query language is SQL (Structured Query Language), which is widely supported in relational databases.	2	K2	CO1								
14.	Mention the types of SQL command SQL commands are divided in to the following categories: 1. Data -definition language 2. Data manipulation language 3. Data Query language 4. Data control language 5. Data administration statements 6. Transaction control statements	2	K1	CO1								
15.	List the SQL domain Types SQL supports the following domain types: Char (n) , varchar (n), int , numeric (p,d) , float(n) , date.	2	K1	CO1								
16.	Give reason for using aggregate functions in SQL queries. Aggregate functions in SQL are used to perform calculations on multiple rows and return a single summarized value. They help in analyzing data efficiently, such as finding totals (SUM), averages (AVG), counts (COUNT), minimum (MIN), and maximum (MAX).	2	K2	CO1								
17.	<table><tr><td colspan="2">Differentiate static and dynamic SQL.</td></tr><tr><td>Static SQL</td><td>Dynamic SQL</td></tr><tr><td>In Static SQL, how database will be accessed is predetermined in the embedded SQL statement.</td><td>In Dynamic SQL, how database will be accessed is determined at run time.</td></tr><tr><td>SQL statements are compiled at compile time.</td><td>SQL statements are compiled at run time.</td></tr></table>	Differentiate static and dynamic SQL.		Static SQL	Dynamic SQL	In Static SQL, how database will be accessed is predetermined in the embedded SQL statement.	In Dynamic SQL, how database will be accessed is determined at run time.	SQL statements are compiled at compile time.	SQL statements are compiled at run time.	2	K2	CO1
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	<table><tr><td>Parsing, Validation, Optimization and Generation of application plan are done at compile time.</td><td>Parsing, Validation, Optimization and Generation of application plan are done at run time.</td></tr><tr><td>It is generally used for situations where data is distributed uniformly.</td><td>It is generally used for situations where data is distributed non uniformly.</td></tr></table>	Parsing, Validation, Optimization and Generation of application plan are done at compile time.	Parsing, Validation, Optimization and Generation of application plan are done at run time.	It is generally used for situations where data is distributed uniformly.	It is generally used for situations where data is distributed non uniformly.			
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It is generally used for situations where data is distributed uniformly.	It is generally used for situations where data is distributed non uniformly.							
18.	Why does SQL allow duplicate tuples in a table or in a query result? If key constraint is not set on a relation every result in a relation will be considered as a tuple and hence SQL allows duplicate tuples in a table. Distinct keyword is used to avoid duplicate tuples in the result.	2	K2	CO1				
19.	List the DDL, DML, DCL and TCL commands. DDL COMMANDS: ● Create ● Alter ● Add ● Modify ● Drop ● Rename ● Drop DML Commands: ● Insert ● Select ● Update ● Delete DCL commands ● Grant -Provide access privilege to user ● Revoke -Get back access privilege from user TCL commands ● Commit ● Rollback Save point	2	K1	CO1				
20.	How does embedded SQL work?Give a sample syntax Embedded SQL works by incorporating SQL statements within a host programming language (such as C or Java). These SQL statements are processed by a precompiler and executed at runtime to interact with the database and retrieve or modify data. Sample :	2	K2	CO1				



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	EXEC SQL BEGIN DECLARE SECTION; int id; char name[20]; EXEC SQL END DECLARE SECTION; EXEC SQL SELECT name INTO :name FROM STUDENT WHERE id = :id;			
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PART B

Q.No.	Questions	Marks	BT L	CO														
1.	Explain the basic architecture of a Database Management System with a neat block diagram	16	K2	CO1														
2.	Discuss in detail about the various Data Models available for data structuring with suitable examples.	16	K2	CO1														
3.	Describe fundamental relational algebra operations and illustrate each with examples	16	K2	CO1														
4.	Demonstrate with necessary syntax and examples the use of DDL, DML, DCL, and TCL commands .	16	K3	CO1														
5.	Consider a student registration database comprising of the below given table schema. Student File <table><tr><td>Student Number</td><td>Student Name</td><td>Address</td><td>Telephone</td></tr></table> Course File <table><tr><td>Course Number</td><td>Description</td><td>Hours</td><td>Professor Number</td></tr></table> Professor File <table><tr><td>Professor Number</td><td>Name</td><td>Office</td></tr></table> Registration file <table><tr><td>Student Number</td><td>Course Number</td><td>Date</td></tr></table> Consider a suitable sample of tuples/records for the above mentioned tables and write DML statements (SQL) and relational algebra equations to solve the queries listed below.	Student Number	Student Name	Address	Telephone	Course Number	Description	Hours	Professor Number	Professor Number	Name	Office	Student Number	Course Number	Date	16	K3	CO1
Student Number	Student Name	Address	Telephone															
Course Number	Description	Hours	Professor Number															
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	1. Which courses does a specific professor teach? and where is his/her office? 2. For a specific student number, in which courses is the student registered and what is his/her name? 3. Who are the professors for a specific student? 4. Who is the student registered in a specific course?			
6.	Assume the following table. • Degree(degcode, name, subject) • Candidate(seatno, degcode, semester, month, year, result) • Marks(seatno, degcode, semester, month, year, papcode, marks) Note: Degcode-degree code, Name-name of the degree (MSc, M.com) Subject-subject of the course. E.g. Phy, Papcode-Paper code E.g. A1 Solve the following queries using SQL: (i) Write a SELECT statement to display all the degree codes which are there in the candidate table but not present in degree table in the order of degcode. (ii) Write a SELECT statement to display the name of all the candidates who have got less than 40 marks in exactly 2 subjects. (iii) Write a SELECT statement to display the name, subject and number of candidates for all degrees in which there are less than 5 candidates. (iv) Write a SELECT statement to display the names of all the candidates who have got highest total marks in MSc.,(Maths)	16	K3	CO1
7.	Demonstrate how Embedded SQL and Dynamic SQL works with necessary examples.	16	K3	CO1
8.	Illustrate the advanced features of SQL with suitable examples.	16	K2	CO1

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Course Code	:	JCS2401
Year / Semester / Branch		II/IV/CSE
Unit No	:	2
Unit Title	:	DESIGN AND NORMALIZATION
No. of Hours	:	9
Weightage in Exam (%)	:	20 %

PART A

Q.No.	Questions	Marks	BTL	CO				
1.	Define normalization. Database normalization is the process of organizing tables and attributes using normal forms to reduce redundancy and improve data integrity. It ensures proper dependency enforcement through constraints and is achieved by either creating a new design (synthesis) or refining an existing one (decomposition).	2	K1	CO2				
2.	Define BCNF. A relational schema is in BCNF if all the redundancy based on functional dependency has been removed, although other types of redundancy may still exist. A relational schema R is in Boyce–Codd normal form if and only if for every one of its dependencies $X \rightarrow Y$, at least one of the following conditions hold: $X \rightarrow Y$ is a trivial functional dependency ($Y \subseteq X$) X is a superkey for schema R	2	K1	CO2				
3.	Compare Fifth Normal Form (5NF) with Fourth Normal Form (4NF). <table border="1"><tr><th>Fourth Normal Form (4NF)</th><th>Fifth Normal Form (5NF)</th></tr><tr><td>Eliminates Multivalued Dependencies (MVDs)</td><td>Eliminates Join Dependencies (JDs)</td></tr></table>	Fourth Normal Form (4NF)	Fifth Normal Form (5NF)	Eliminates Multivalued Dependencies (MVDs)	Eliminates Join Dependencies (JDs)			
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	<table> <tr> <td>Deals with independent multiple values in a relation</td> <td>Deals with complex relationships requiring decomposition into smaller tables</td> </tr> <tr> <td>A relation is in 4NF if it has no non-trivial MVDs</td> <td>A relation is in 5NF if every join dependency is implied by candidate keys</td> </tr> <tr> <td>Removes redundancy caused by multivalued attributes</td> <td>Removes redundancy caused by improper decomposition</td> </tr> <tr> <td>Easier to understand and apply</td> <td>More complex and rarely required in practice</td> </tr> <tr> <td>Splits tables to remove MVDs</td> <td>Further decomposes tables to ensure lossless join without redundancy</td> </tr> <tr> <td>Common in practical database design</td> <td>Rare, used in highly complex databases</td> </tr> </table>	Deals with independent multiple values in a relation	Deals with complex relationships requiring decomposition into smaller tables	A relation is in 4NF if it has no non-trivial MVDs	A relation is in 5NF if every join dependency is implied by candidate keys	Removes redundancy caused by multivalued attributes	Removes redundancy caused by improper decomposition	Easier to understand and apply	More complex and rarely required in practice	Splits tables to remove MVDs	Further decomposes tables to ensure lossless join without redundancy	Common in practical database design	Rare, used in highly complex databases	2	K2	CO2
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4.	Mention the common pitfalls in relational database design. • Data redundancy • Insertion anomaly • Deletion anomaly • Update anomaly • Poor normalization • Lack of data integrity • Improper selection of keys	2	K1	CO2												
5.	Classify and define the types of participation constraints. Total: The participation of an entity set E in a relationship set R is said to be total if every entity in E participates in at least one relationship in R. Partial: if only some entities in E participate in relationships in R, the participation of entity set E in relationship R is said to be partial participation.	2	K2	CO2												
6.	Why is lossless decomposition important for normalization? Decomposition must be lossless. It means that the information should not get lost from the relation that is decomposed. It gives a guarantee that the join will result in the same relation as it was decomposed. Example: Let's take 'E' is the Relational Schema, With instance 'e'; is decomposed into: E1, E2, E3, En; With instance: e1, e2, en, If $e1 \bowtie e2 \bowtie e3 \bowtie en$, then it is called as 'Lossless Join Decomposition'.	2	K2	CO2												



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7.	List the various types of attributes in ER model. • Simple attribute. • Composite attribute. • Derived attribute. • Single-value attribute • Multi-value attribute .	2	K1	CO2
8.	Outline the concept of multi-valued dependency and its characteristics. When existence of one or more rows in a table implies one or more other rows in the same table, then the Multi-valued dependencies occur. If a table has attributes P, Q and R, then Q and R are multi-valued facts of P. It is represented by double arrow: $\twoheadrightarrow$ For example: $P \twoheadrightarrow Q$ $P \twoheadrightarrow R$ In the above case, Multivalued Dependency exists only if Q and R are independent attributes. A table with multivalued dependency violates the 4NF.	2	K2	CO2
9.	State the conditions for lossless decomposition. Suppose that we decompose the schema $R=(A,B,CD,E)$ into (A,B,C) and (A,D,E) , show that this decomposition is a lossless decomposition if the following set F of functional dependencies holds: $A \rightarrow BC \quad CD \rightarrow E \quad B \rightarrow D \quad E \rightarrow A$ Decomposition is a method used to divide large schema R into number of small relation schemas $R_1, R_2 \dots R_n$ in such a way that $R_1 \cup R_2 \cup \dots \cup R_n = R$ The decomposition is lossless if either of the following holds: $R_1 \cap R_2 \rightarrow R_1$ $R_1 \cap R_2 \rightarrow R_2$ Here $R_1 \cap R_2$ is A which is common attribute and A is a candidate Key. Therefore, the given decomposition is lossless decomposition.	2	K2	CO2
10.	Why is 4NF more desirable than BCNF? • 4NF is more restrictive than BCNF • 4NF reduces repetition of information than BCNF • Every 4NF schema is in BCNF, but there are BCNF schemas that are not in 4NF	2	K2	CO2
11.	Summarize the issues to be considered while developing an ER diagram. • Use of entity sets Vs attributes. • Use of entity sets Vs relationship sets.	2	K2	CO2



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	• Binary Vs N-ary relationship sets. • Placement of relationship attributes.									
12.	List out the extended ER features. • Specialization • Generalization • Attribute Inheritance • Constraints on Generalizations • Aggregation • Reduction to Relation Schemas	2	K1	CO2						
13.	Interpret the concept of functional dependency in DBMS. A functional dependency, denoted by $X \rightarrow Y$, between two sets of attributes X and Y that are subsets of R specifies a constraint on the possible tuples that can form a relation state r of R. The constraint is that, for any two tuples t1 and t2 in r that have $t1[X] = t2[X]$, they must also have $t1[Y] = t2[Y]$.	2	K2	CO2						
14.	List and define the various forms of trivial dependency • Trivial – If a functional dependency (FD) $X \rightarrow Y$ holds, where Y is a subset of X, then it is called a trivial FD. Trivial FDs always hold. • Non-trivial – If an FD $X \rightarrow Y$ holds, where Y is not a subset of X, then it is called a non-trivial FD. • Completely non-trivial – If an FD $X \rightarrow Y$ holds, where x intersects $Y = \Phi$, it is said to be a completely non-trivial FD.	2	K1	CO2						
15.	Mention the reasons for the occurrence of trivial dependencies in relational schemas. • A trivial functional dependency occurs when the dependent attribute is a subset of the determinant (e.g., $A \rightarrow A$); this naturally arises from the definition of functional dependency. • They exist due to reflexivity rule in normalization, ensuring every attribute functionally determines itself, which helps maintain logical consistency in schemas.	2	K1	CO2						
16.	Differentiate Prime and Non-prime attributes. <table><tr><th>Prime Attribute</th><th>Non-Prime Attribute</th></tr><tr><td>An attribute that is part of any candidate key</td><td>An attribute that is not part of any candidate key</td></tr><tr><td>Helps uniquely identify tuples (records)</td><td>Does not uniquely identify tuples</td></tr></table>	Prime Attribute	Non-Prime Attribute	An attribute that is part of any candidate key	An attribute that is not part of any candidate key	Helps uniquely identify tuples (records)	Does not uniquely identify tuples	2	K2	CO2
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Crucial for normalization and key formation	Used for storing additional Information									
Usually appears in functional dependencies involving keys	Often depends on prime attributes									
Present in candidate/primary key	Not present in any key									
17.	Mention the use of a Primary Key . Primary key is a set of one or more attributes that can uniquely identify record from the relation; it will not accept null values and redundant values. A relation can have only one primary key.	2	K1	CO2						
18.	Define super key in DBMS. A super key is a set of one or more attributes that collectively allows us to identify uniquely an entity in the entity set.	2	K1	CO2						
19.	Discuss Domain-Key Normal Form (DKNF) and state the conditions under which a relational schema satisfies DKNF A relation schema is said to be in DKNF only if all the constraints and dependencies that should hold on the valid relation state can be enforced simply by enforcing the domain constraints and the key constraints on the relation.	2	K2	CO2						
20.	Why is it important to eliminate transitive dependencies? Transitive dependency is a functional dependency which holds by virtue of transitivity. A transitive dependency can occur only in a relation that has three or more attributes. Let A, B, and C designate three distinct attributes (or distinct collections of attributes) in the relation. Suppose all three of the following conditions hold: 1.A-> B 2. It is not the case that B->A 3. B->C Then the functional dependency A->C (Which follows from 1 and 3 by the axiom of transitivity) is a transitive dependency.	2	K2	CO2						

PART B

Q.No.	Questions	Marks	BTL	CO
1.	Discuss in detail about Mapping cardinalities and Enhanced Entity–Relationship (EER).	16	K2	CO2
2.	Explain the Entity–Relationship (ER) model by taking a case study and describe its entities, attributes, relationships, and			CO2



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	constraints. Sample Case Studies: Hospital Management System (or) Railway Ticket Reservation System (or) Library Management System.	16	K2	
3.	Construct an E-R Diagram for the "Restaurant Menu Ordering System", which will facilitate the food items ordering and services within a restaurant. The entire restaurant scenario is detailed as follows. The Customer is able to view the food items menu, call the waiter, place orders and obtain the final bill through the computer kept in their table. The waiters through their wireless tablet PC are able to initialize a table for customers, control the table functions to assist customers, orders, send orders to food preparation staff (chef) and finalize the customer's bill. The food preparation staffs (Chefs), with their touch-display interface to the system, are able to view orders sent to the kitchen by waiters. During preparation, they are able to let the waiter know the status of each item, and can send notification when items are completed. The system should have full accountability and logging facilities, and should support supervisor actions to account for exceptional circumstances, such as a meal being refunded or walked out on.	16	K3	CO2
4.	Apply the ER-to-Relational mapping process to convert the given ER diagram into a relational schema. Illustrate each step involved in the mapping, including handling of strong entities, weak entities, relationships (1:1, 1:N, M:N), multivalued attributes, and recursive relationships. Present the final relational schema with appropriate primary and foreign keys.	16	K3	CO2



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5.	Explain each of the following normal forms with example. (i) First Normal Form (ii) Second Normal Form (iii) Third Normal Form (iv) Boyce-Codd Normal Form	16	K2	CO2
6.	Demonstrate the usage of the following Normal forms in eliminating redundancy and anomalies: (i) Fourth Normal Form (4NF) (ii) Multivalued Dependency (MVD) (iii) Fifth Normal Form (5NF) (iv) Join Dependency and Lossless Decomposition	16	K3	CO2
7.	Consider the relation: STUDENT_COURSE (StudentID, StudentName, CourseID, CourseName, Instructor, InstructorPhone) • Identify the candidate key(s) for the relation. • Determine the highest normal form of the given relation (1NF, 2NF, or 3NF). • Normalize the relation up to 3NF, showing all steps clearly. • Decompose the relation into smaller tables and specify primary keys and foreign keys.	16	K3	CO2



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8.	Illustrate full, partial, and transitive dependencies using a sample table.	16	K2	CO2
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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Regulation	:	2023
Course Name	:	DATABASE MANAGEMENT SYSTEMS
Course Code	:	JCS2401
Year / Semester / Branch	:	II / IV / CSE
Unit No	:	3
Unit Title	:	TRANSACTION PROCESSING
No. of Hours	:	9
Weightage in Exam (%)	:	20%

QUESTION BANK

PART A

Q.No.	Questions	Marks	BTL	CO
1.	Define transaction. • A transaction is a set of operations executed as a single logical unit that accesses and may modify database data. • It follows the all-or-nothing rule and ensures data integrity and consistency during execution.	2	K1	CO3
2.	List out the various states of a transaction. Give the diagrammatic representation of the same. Active – the initial state; the transaction stays in this state while it is executing Partially committed – after the final statement has been executed. Failed - after the discovery that normal execution can no longer proceed. Aborted – after the transaction has been rolled back and the database restored to its state prior to the start of the transaction. Committed – after successful completion	2	K1	CO3



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	<pre> graph LR Active((Active)) --> PartiallyCommitted((Partially Committed)) Active --> Failed((Failed)) PartiallyCommitted --> Committed((Committed)) Failed --> Aborted((Aborted)) </pre>			
3.	How do schedules ensure consistency when managing concurrent transactions in DBMS? Schedule – a sequence of instructions that specifies the chronological order in which instructions of concurrent transactions are executed. A schedule for a set of transactions must consist of all instructions of those transactions must preserve the order in which the instructions appear in each transaction.	2	K2	CO3
4.	Identify the function of serializability? Give its various forms. Serializability ensures that the execution of multiple transactions gives the same result as if they were executed one after another (serially). Its main function is to maintain data consistency in the database. It helps in avoiding conflicts when transactions run at the same time. It also ensures that the database remains correct and reliable after concurrent execution. Different forms of schedule equivalence give rise to the notions of: • Conflict serializability • View serializability	2	K2	CO3
5.	Illustrate when two instructions are said to be in conflict in transaction? Instructions li and lj of transactions Ti and Tj, respectively, conflict if and only if there exists some item Q accessed by both li and lj, and at least one of these instructions wrote Q. • $li = \text{read}(Q)$, $lj = \text{read}(Q)$. li and lj don't conflict. • $li = \text{read}(Q)$, $lj = \text{write}(Q)$. They conflict. • $li = \text{write}(Q)$, $lj = \text{read}(Q)$. They conflict • $li = \text{write}(Q)$, $lj = \text{write}(Q)$. They conflict	2	K2	CO3
6.	State the property of conflict serializability. If a schedule S can be transformed into a schedule S' by a series of swaps of non-conflicting instructions, we say that S and S' are	2	K1	CO3



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	conflict equivalent and that schedule S is conflict serializable if it is conflict equivalent to a serial schedule			
7.	Outline the conditions for view serializability. - Let S and S' be two schedules with the same set of transactions. S and S' are view equivalent if the following three conditions are met, for each data item Q, - If in schedule S, transaction T_i reads the initial value of Q, then in schedule S' also transaction T_i must read the initial value of Q. - If in schedule S transaction T_i executes $\text{read}(Q)$, and that value was produced by transaction T_j (if any), then in schedule S' also transaction T_i must read the value of Q that was produced by the same $\text{write}(Q)$ operation of transaction T_j. - The transaction (if any) that performs the final $\text{write}(Q)$ operation in schedule S must also perform the final $\text{write}(Q)$ operation in schedule S'.	2	K2	CO3
8.	Identify the different modes of lock in a transaction. Lock is the most commonly used to implement the requirement is to allow a transaction to access a data item only if it is currently holding a lock on that item. Data items can be locked in two modes: - Exclusive (X) mode. Data items can be both read as well as written. X-lock is requested using lock-X instruction. - Shared (S) mode. Data item can only be read. S-lock is requested using lock-S instruction. Lock requests are made to concurrency-control manager. Transaction can proceed only after request is granted.	2	K1	CO3
9.	How does a transaction locking protocol work ? - A transaction locking protocol controls access to data by assigning locks (shared or exclusive) before operations are performed. - It ensures concurrent transactions do not interfere, maintaining data consistency by allowing only safe access and releasing locks after execution..	2	K2	CO3
10.	Why are ACID properties crucial for transaction management in databases?	2	K2	CO3



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	● Atomicity - Atomicity means that an entire transaction either takes place all at once or it doesn't occur at all. It means that there's no midway ● Consistency - Consistency means that we have to maintain the integrity constraints so that any given database stays consistent both before and after a transaction. ● Isolation - Isolation ensures the occurrence of multiple transactions concurrently without a database state leading to a state of inconsistency ● Durability - The durability property states that once the execution of a transaction is completed, the modifications and updates on the database gets written on and stored in the disk.			
11.	Classify the phases of two-phase locking protocol. Growing phase: a transaction may obtain locks but not release any lock. Shrinking phase: a transaction may release locks but may not obtain any new locks.	2	K2	CO3
12.	Define Deadlock with an example. A deadlock occurs when two or more tasks permanently block each other by each task having a lock on a resource that the other tasks are trying to lock. For example, let us assume we have two processes, P1 and P2. Now, process P1 is holding the resource R1 and is waiting for the resource R2. At the same time, process P2 is having the resource R2 and is waiting for the resource R1.	2	K1	CO3
13.	Differentiate between the strict two-phase locking protocol and the rigorous two-phase locking protocol. ● Strict two-phase locking: □ It holds all it is exclusive that is write locks until commit time, one time granted. □ It locks all its items before it starts thus once transaction starts, it is in its shrinking phase ● Rigorous two-phase locking:	2	K2	CO3



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	□ It is more restrictive as compared to standard 2PL in that it enforces the rule that no locks can be released through a transaction until after it commits or aborts. □ It does not unlock any of its items until after it terminates to the transaction is in its expanding phase till it ends.			
14.	Interpret the working of Save point and Rollback commands in SQL. A Save point is a logical rollback point within a transaction. When you set a save point, whenever an error occurs past a save point, you can undo the events you have done up to the save point using the rollback. Syntax: begin; savepoint <savepoint name>; Rollback command serves only in the creation of a SAVEPOINT among all the transactional statements. The ROLLBACK command is used to undo a group of transactions. Syntax : Rollback to <savepoint _name>;	2	K2	CO3
15.	Specify different isolation levels in the database ● Serializable. ● Repeatable reads. ● Read committed. ● Read uncommitted. ● Dirty reads. ● Non-repeatable reads.	2	K2	CO3
16.	List and define the two approaches for deadlock prevention. ● Wait–Die Scheme: A non-pre-emptive technique based on timestamps where an older transaction is allowed to wait for a younger one, but a younger transaction requesting a resource held by an older one is aborted (dies) and restarted later. ● Wound–Wait Scheme: A pre-emptive technique where an older transaction pre-empt (wounds) a younger one, forcing the younger transaction to abort, while a younger transaction waits if the resource is held by an older one.	2	K1	CO3



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17.	Why is a wait-for <i>graph</i> important in Deadlock? The wait-for graph is crucial for detecting and resolving deadlocks efficiently, ensuring smooth transaction processing and maintaining database consistency.	2	K2	CO3
18.	Name the two operations involved in recovery • Undo (TRX): This command is used to restore all records updated by transactions to the old value. • Redo (TRX): This command is used to set the value of all records updated by a transaction to the new value.	2	K1	CO3
19.	List the four conditions for deadlock to occur in a system. • mutual exclusion • hold and wait or partial allocation • no pre-emption • resource waiting or circular wait	2	K1	CO3
20.	In a multi-user database system, how does isolation prevent issues like dirty reads, non-repeatable reads, and phantom reads? Transaction isolation is an important part of any transactional system. It deals with consistency and completeness of data retrieved by queries un affecting a user data by other user actions. A database acquires locks on data to maintain a high level of isolation.	2	K2	CO3

PART B

Q.No.	Questions	Marks	BTL	CO
1.	i) Consider the following schedules. The actions are listed in the order they are scheduled, and pre-fixed with the transaction name. S1: T1: R(X), T2: R(X), T1: W(Y), T2: W(Y), T1: R(X), T2: R(Y) S2: T3: W(X), T1: R(X), T1: W(Y), T2: R(Z), T2: W(Z), T3: R(Z) For each of the schedules, analyse the following questions: (a) Is the schedule conflict serializable? If so, what are all the conflict equivalent serial schedules? (b) Is the schedule View serializable? If so, what are all the view equivalent serial schedules?	8	K4	CO3



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	ii) Consider the following two transactions: <i>T1</i>: read(<i>A</i>); read(<i>B</i>); if <i>A</i> = 0 then <i>B</i> := <i>B</i> + 1; write(<i>B</i>). <i>T2</i>: read(<i>B</i>); read(<i>A</i>); if <i>B</i> = 0 then <i>A</i> := <i>A</i> + 1; write(<i>A</i>). Add lock and unlock instructions to transactions <i>T1</i> and <i>T2</i>, so that they observe the two-phase locking protocol. Analyse if the execution of these transactions result in a deadlock?	8		
2.	Illustrate the process of Two-Phase Locking Protocol to a fund transfer transaction between two bank accounts.	16	K2	CO3
3.	i) Two sellers update the same account balance of ₹10,000 concurrently: • <i>T1</i> withdraws ₹2,000 • <i>T2</i> withdraws ₹3,000 Evaluate how the lost update problem occurs in this scenario with example. ii) Transaction <i>T1</i> updates a product price from ₹500 to ₹450 but has not committed yet. Transaction <i>T2</i> reads this uncommitted value and displays it to customers. Later, <i>T1</i> is rolled back. Assess the consequences of dirty reads on database consistency and reliability with example.	8 8	K5	CO3
4.	Analyse how the ARIES recovery algorithm handles a crash scenario.	16	K4	CO3
5.	Explain how deadlocks can be managed in a high-concurrency database using both detection and prevention techniques.	16	K2	CO3
6.	Evaluate the effectiveness of SQL facilities for managing Concurrency Control and Database Recovery in ensuring reliable and efficient transaction processing.	16	K5	CO3



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QUESTION BANK

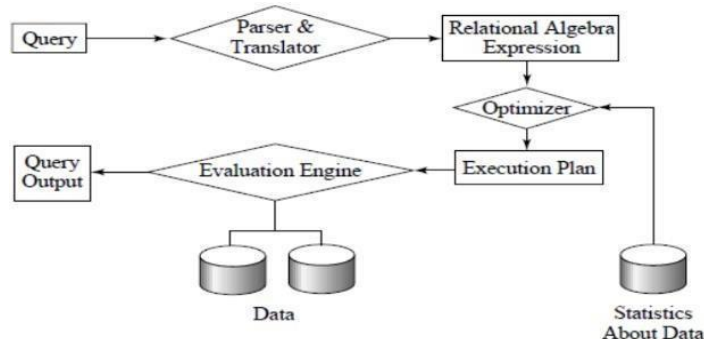
Regulation	:	2023
Course Name	:	DATABASE MANAGEMENT SYSTEMS
Course Code	:	JCS2401
Year / Semester / Branch	:	II / IV / CSE
Unit No	:	4
Unit Title	:	DATA ORGANIZATION AND QUERY OPTIMIZATION
No. of Hours	:	9
Weightage in Exam (%)	:	20%

PART A

Q.No.	Questions	Marks	BTL	CO
1.	Define the term RAID. RAID (Redundant Array of Independent Disks) is a technology that combines multiple physical disks into a single logical unit to improve performance, reliability, or both.	2	K1	CO4
2.	Outline the factors to be considered when choosing a RAID level • Monetary cost of extra disk storage requirements. • Performance requirements in terms of number of I/O operations • Performance when a disk has failed and Performances during rebuild	2	K2	CO4
3.	Define a record in a file. A record in a file is a collection of related fields that are treated as a single unit, representing a complete set of information about one specific entity. It is a fundamental data structure component, where multiple records are stored sequentially to form a file.	2	K1	CO4
4.	How does indexing work ? Indexing is a data structure technique used to speed up data retrieval by creating a separate, sorted structure that allows the	2	K2	CO4

	database to locate rows without scanning the entire table. It dramatically reduces disk I/O, enhances query performance and is crucial for handling large datasets.			
5.	List out the types of storage devices Primary storage, Secondary storage, Tertiary storage, Volatile storage, Nonvolatile storage etc.	2	K1	CO4
6.	Define an ordered index in DBMS. An ordered index in a Database Management System (DBMS) is a specialized data structure, typically a sorted list of key-value pairs (search key and pointer), that stores index entries in a sorted order to facilitate fast data retrieval. Ordered indexes enhance query performance by allowing binary search algorithms, significantly reducing the need for full table scans.	2	K1	CO4
7.	Mention the structure of B+ tree. A B+ tree is a self-balancing, multi-level tree data structure used in DBMS for indexing, optimizing search and range queries by storing all data/pointers exclusively in leaf nodes. Leaf nodes are linked sequentially for efficient traversal, while internal nodes only store keys to guide search, making them ideal for high-fanout disk-based storage.	2	K1	CO4
8.	Define a B-tree. A B-tree is a self-balancing, m-way tree data structure used in DBMS to optimize searching, insertion, and deletion of large datasets, specifically designed to minimize disk I/O operations. It maintains sorted data with all leaf nodes at the same depth, ensuring efficiency for operations while keeping the tree shallow.	2	K1	CO4
9.	Specify the function of static hashing. Static hashing is a data storage technique where a fixed number of buckets (memory locations) are used, and a consistent hash function maps search keys to specific bucket addresses. The number of buckets remains constant, making it ideal for scenarios with static, predictable data volume.	2	K1	CO4
10.	State the process of query optimization Query optimization in a Database Management System (DBMS) is the process of selecting the most efficient execution plan for a given SQL query from among all possible equivalent plans. The primary goal is to minimize the resources used (such as CPU, memory, and disk I/O) and the overall query execution	2	K1	CO4

	time to provide faster results to users.			
11.	Why is RAID important for file organization? RAID (Redundant Array of Independent Disks) is used in DBMS to improve data reliability, increase storage performance (I/O speed), and ensure high availability by distributing data across multiple disks. It protects against data loss via mirroring or parity, allowing databases to continue running despite hardware failures.	2	K2	CO4
12.	Interpret the working of sequential file organization. Sequential file organization is a fundamental method where records are stored and accessed in a linear, one-after-another sequence. This method is simple in design and particularly effective for processes that require accessing most or all records in order, such as generating payroll slips or reports.	2	K2	CO4
13.	Differentiate between Primary Index and Secondary Index: • Primary Index: □ Created on the primary key of a file. □ Data file is physically ordered based on this key. □ Only one primary index is possible. □ Usually a sparse index. • Secondary Index: □ Created on a non-key field. □ Data file is not ordered based on this field. □ Multiple secondary indexes can exist. □ Usually a dense index.	2	K2	CO4
14.	How does hashing improve search efficiency? Hashing improves search efficiency in a DBMS by providing direct, constant-time ($\Theta(1)$ on average) access to data records, eliminating the need for sequential searches or tree traversals, which are significantly slower for large databases.	2	K2	CO4
15.	Give reasons for the usage of ordered indices. • Fast Data Retrieval (Search Speed). • Efficient Range Queries	2	K2	CO4

	- Reduced Disk I/O - Optimized Sorting and Grouping - Support for Unique Constraints			
16.	How does Linear probing work? - Linear probing is an open addressing collision resolution technique where, if a hash collision occurs, the system searches for the next available slot sequentially in the hash table. - If a key hashes to an occupied position, the system checks the next slot using: $h_i(k) = (h(k) + i) \mod m$ where i = 0,1,2,3,...and m is the table size.	2	K2	CO4
17.	Interpret the limitation of static hashing The primary limitation of static hashing in DBMS is its fixed number of buckets, which cannot adapt to changing database sizes. As data grows, this causes frequent bucket overflows (collisions), leading to long overflow chains and degraded search performance, while shrinking data leads to wasted memory space.	2	K2	CO4
18.	Illustrate on dynamic hashing Dynamic hashing adjusts the number of buckets, growing or shrinking to accommodate data changes, which improves performance for dynamic datasets and minimizes overflow.	2	K2	CO4
19.	List the steps in in Query Processing and give a diagrammatic illustration . Basic steps in query processing - Parsing and translation - Optimization - Evaluation  <pre> graph TD Query[Query] --> PT{Parser & Translator} PT --> RA[Relational Algebra Expression] RA --> Opt{Optimizer} Opt --> EP[Execution Plan] EP --> EE{Evaluation Engine} EE --> QO[Query Output] Data[(Data)] --> EE Stats[(Statistics About Data)] --> EE </pre>	2	K2	CO4

20.	Why is query optimization important? Query optimization in a DBMS is crucial because it ensures that queries run efficiently by selecting the most cost-effective execution plan out of many possibilities. This process minimizes the use of system resources, resulting in faster data retrieval and overall better database performance.	2	K2	CO4
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PART B

Q.No.	Questions	Marks	BTL	CO
1	Discuss in detail how records are stored and accessed using secondary storage devices, with suitable examples.	16	K2	CO4
2	Describe the usage of RAID techniques in a storage system.	16	K2	CO4
3	Categorize the various Indexing and Hashing techniques used for file organization and explain each with a suitable example.	16	K4	CO4
4	Analyze the structure and operations of B-tree and B+ tree indexing techniques. Illustrate their differences with suitable examples.	16	K4	CO4
5	Explain in detail the different hashing techniques suitable method for efficient data retrieval.	16	K2	CO4
6	Illustrate the steps involved in query processing and describe how each step improves query performance.	16	K2	CO4

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QUESTION BANK

Regulation	:	2023
Course Name	:	DATABASE MANAGEMENT SYSTEMS
Course Code	:	JCS2401
Year / Semester / Branch	:	II / IV / CSE
Unit No	:	5
Unit Title	:	ADVANCED DATABASES
No. of Hours	:	9
Weightage in Exam (%)	:	20%

PART A

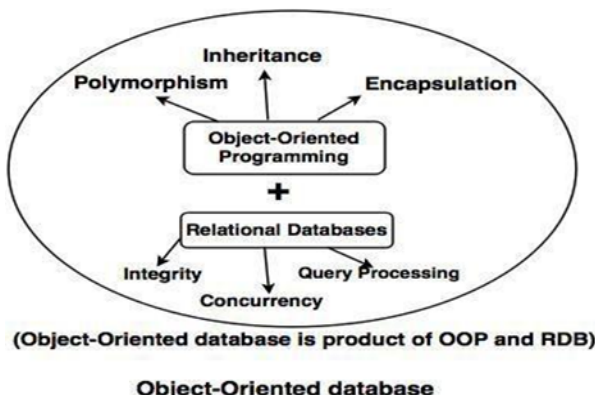
Q.No.	Questions	Marks	BTL	CO
1	Define Object-Oriented Database (OODBMS). Answer: An Object-Oriented Database Management System (OODBMS) is a database system that stores, manages, and retrieves data in the form of objects, as used in object-oriented programming (OOP). It integrates database capabilities with object-oriented programming features such as classes, objects, inheritance, polymorphism, and encapsulation.	2	K1	CO5
2	List the advantages of OODB • Supports complex data types (e.g., multimedia, images, videos). • Encapsulation improves data security and integrity. • Inheritance enables reuse of data and reduces redundancy. • Better modeling of real-world entities using objects. • Seamless integration with object-oriented programming languages. • Reduced impedance mismatch between application and database.	2	K1	CO5
3	Outline the aspects of Objects and Literals. Objects are entities that have identity, state (attributes), and behavior (methods). Literals are constant values that do not have identity and are	2	K2	CO5



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	used to represent data directly.															
4	Define ODL. ODL language is used to create object specifications: classes and interfaces-Using the specific language bindings to specify how ODL constructs can be mapped to constructs in specific programming language, such as C++, SMALLTALK, and JAVA	2	K1	CO5												
5	Give the diagrammatic representation Object Oriented Database.  (Object-Oriented database is product of OOP and RDB) Object-Oriented database	2	K1	CO5												
6	How does an Object-relational database work? An Object-Relational Database combines relational tables with object-oriented features, allowing data to be stored in tables along with complex data types like objects, arrays, and user-defined types. It works by extending SQL to support object concepts such as inheritance, methods, and encapsulation, enabling more flexible and powerful data modeling.	2	K2	CO5												
7	Differentiate between E-R and ODL <table border="1"> <thead> <tr> <th>Feature</th> <th>E/R Model (Entity-Relationship)</th> <th>ODL (Object Definition Language)</th> </tr> </thead> <tbody> <tr> <td>Approach</td> <td>Conceptual data model</td> <td>Object-oriented data model</td> </tr> <tr> <td>Representation</td> <td>Uses entities, attributes, relationships</td> <td>Uses objects, classes, and methods</td> </tr> <tr> <td>Data Types</td> <td>Simple data types</td> <td>Supports complex and user-defined data types</td> </tr> </tbody> </table>	Feature	E/R Model (Entity-Relationship)	ODL (Object Definition Language)	Approach	Conceptual data model	Object-oriented data model	Representation	Uses entities, attributes, relationships	Uses objects, classes, and methods	Data Types	Simple data types	Supports complex and user-defined data types	2	K2	CO5
Feature	E/R Model (Entity-Relationship)	ODL (Object Definition Language)														
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	Behaviour	Does not support methods	Supports methods (behaviour)			
	Inheritance	Not supported	Supported			
	Usage	Used for database design (conceptual level)	Used to define object database schema			
8	Mention the functionality of a XML database. An XML database is a data persistence software system that allows data to be stored in XML format. These data can then be queried, exported and serialized into the desired format. XML databases are usually associated with document-oriented databases			2	K1	CO5
9	Give the Syntax for a basic OQL command. class Person (extent persons key ssn) { } class Faculty extends Person ENTRY POINTS (extent faculty) { } class Department (extent department key dname) { }			2	K1	CO5
10	Define the hierarchical structure of XML document XML documents have a hierarchical structure and can conceptually be interpreted as a tree structure, called an XML tree. XML documents must contain a root element (one that is the parent of all other elements). All elements in an XML document can contain sub elements, text and attributes.			2	K1	CO5
11	Interpret the use of XML Schema XML Schema is commonly known as XML Schema Definition (XSD). It is used to describe and validate the structure and the content of XML data. XML schema defines the elements, attributes and data types. Schema element supports Namespaces.			2	K2	CO5
12	List the components of mobile database. • Mobile Client • Wireless Communication Network • Server (Fixed Host) • Database Management System (DBMS) • Middleware			2	K1	CO5



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13	Differentiate between OODBMS and RDBMS.			2	K2	CO5
	Feature	OODBMS	RDBMS			
	Data Model	Object-oriented (objects & classes)	Relational (tables & rows)			
	Data Representation	Stores data as objects	Stores data in tables			
	Inheritance	Supported	Not supported			
	Query Language	Uses OQL or similar	Uses SQL			
14	Illustrate on DTD. A document type definition (DTD) contains a set of rules that can be used to validate an XML file. After you have created a DTD, you can edit it manually, adding declarations that define elements, attributes, entities, and notations, and how they can be used for any XML files that reference the DTD file			2	K2	CO5
15	Outline the steps to create DTD. 1. Create a project to contain the DTD if needed. 2. In the workbench, click File > New > Other and select XML>DTD. Click Next. 3. Select the project or folder that will contain the DTD. 4. In the File name field, type the name of the DTD, for example MyDTD.dtd. The name of your DTD file must end with the extension.dtd .Click Next 5. Optional: You can use a DTD template as the basis for your new DTD file. To do so, click the Use DTD Template check box, and select the template you want to use. 6. Click Finish.			2	K2	CO5
16	Give reasons for why a XML Schema is better considered to a DTD. • It allows the text that appears in elements to be constrained to specific types, such as numeric types in specific formats or complex types such as sequences of elements of other types. • It allows user-defined types to be created. • It allows uniqueness and foreign-key constraints. • It standardizes object database models and query languages for interoperability.			2	K2	CO5
17	How do mobile databases differ from traditional databases? A mobile database is a database that is designed to operate on mobile devices such as smartphones, tablets, and			2	K2	CO5



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	laptops, allowing offline access and synchronization with a central database. It uses centralized and require a constant network connection, mobile databases support disconnected operations, replication, and synchronization for data consistency.			
18	Distinguish between XML and HTML. XML is used for data storage, while HTML is used for data presentation. XML is Case-sensitive (e.g., <Name> and <name> are different), while HTML is not case-sensitive (e.g., <P> is the same as <p>).	2	K2	CO5
19	Illustrate on temporal database. A temporal database in DBMS is a database system that stores and manages data with a time dimension, enabling the tracking of data changes over time and providing historical analysis capabilities.	2	K2	CO5
20	Why is Spatial Index important? A spatial index is used by a spatial database to optimize spatial queries. Database systems use indices to quickly look up values by sorting data values in a linear (e.g. alphabetical) order. Instead, spatial databases use a <i>spatial</i> index designed specifically for multi-dimensional ordering.	2	K2	CO5

PART B

Q.No.	Questions	Marks	BTL	CO
1	Apply the concept of Object-Oriented Data Model to design a database schema for any system of your choice .	16	K3	CO5
2	Differentiate between Object-Oriented Databases and Object-Relational Databases with suitable examples.	16	K3	CO5
3	Illustrate how an ODMG Object Model is created and its role in standardizing Object-Oriented Databases.	16	K2	CO5
4	Construct an XML document for a student database and demonstrate its structure using elements and attributes.	16	K3	CO5
5	Illustrate the process of creating a XML schema for a given database.	16	K2	CO5
6	Summarize the challenges and advantages of Mobile Databases in distributed environments.	16	K2	CO5
7	Demonstrate how Temporal Database model can be used to manage employee records with time-varying data.	16	K3	CO5
8	Discuss in detail about Spatial Database system for a GIS application.	16	K2	CO5

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